Discrete random variables and the binomial distribution

Lecture and worked examples

Maths Explorer

Reference notes

The expected value is $E(X) = \sum_i x_i p_i$. It represents the long-run average value when the experiment is repeated under the same conditions; it is not necessarily a value actually taken by X.

Reference notes

X follows a binomial distribution $\mathcal{B}(n, p)$ when X counts the number of successes in n independent Bernoulli trials with the same probability of success p. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad E(X) = np.$$

Expected value of a random variable

- Define a distribution.
- Calculate an expected value.
- Interpret a theoretical average.

Expected value of a random variable

A game costs 2 euros. The gross winnings are 0, 3 or 10 euros with probabilities 0.6, 0.3, 0.1. Construct the distribution of the net gain X, calculate $E(X)$ and decide whether the game is favourable to the player.

Expected value of a random variable

The distribution of X assigns to each value x_i the probability p_i. The expected value $E(X) = \sum x_i p_i$ is a theoretical long-run average.

Expected value of a random variable

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Expected value of a random variable

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Binomial distribution: recognising a situation

- Recognise a Bernoulli scheme.
- Identify n and p.
- Define X.

Binomial distribution: recognising a situation

A customer accepts an offer with probability 0.25. An adviser contacts eight customers independently. Define X, justify the binomial distribution and identify n and p.

Binomial distribution: recognising a situation

$X \sim \mathcal{B}(n, p)$ if X counts the successes in n independent trials with the same probability p.

Binomial distribution: recognising a situation

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Binomial distribution: recognising a situation

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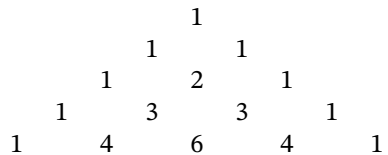
Pascal’s triangle and binomial coefficients

- Build Pascal’s triangle.
- Read $\binom{n}{k}$.
- Interpret a coefficient.

Pascal's triangle and binomial coefficients

Build rows $n = 0$ to $n = 6$ of Pascal's triangle. Explain the calculation of the number 15 in row 6, then read $\binom{6}{2}$ and $\binom{6}{4}$.



Pascal's triangle and binomial coefficients

Pascal's triangle satisfies $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. The edges equal 1.

Pascal’s triangle and binomial coefficients

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Pascal’s triangle and binomial coefficients

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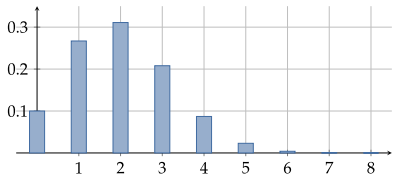
Binomial probabilities

- Calculate $P(X = k)$.
- Translate at least/at most.
- Use the complement.

Binomial probabilities

For $X \sim \mathcal{B}(8, 0.25)$, calculate $P(X = 2)$, $P(X = 0)$, then $P(X \geq 1)$. Translate each event into a sentence.



Binomial probabilities

If $X \sim \mathcal{B}(n, p)$, then $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Translate “at least”, “at most” and “none” carefully.

Binomial probabilities

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Binomial probabilities

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

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Chapter review and test

- Calculate and interpret $E(X)$.
- Solve a binomial problem.
- Analyse one's mistakes.

Chapter review and test

A centre receives 120 groups of 10 customers, each customer complaining with probability 0.08. Estimate the average number of complaints per group, then over all the groups. Discuss what the expected value does and does not allow us to predict.

Chapter review and test

For $X \sim \mathcal{B}(n, p)$, $E(X) = np$. The value gives the average number of successes over a large number of series of n trials.

Chapter review and test

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

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